D1 Binomial expansion

Name: ________________

Class: ________________

Date: ________________

Time: **85 minutes**

Marks: **70 marks**

Comments:

Q1.

(a) (i) Find the binomial expansion of $(1 + x)^{-1}$ up to the term in x^3.

(2)

(ii) Hence, or otherwise, obtain the binomial expansion of $\frac{1}{1+3x}$ up to the term in x^3.

(2)

(b) Express $\frac{1+4x}{(1+x)(1+3x)}$ in partial fractions.

(3)

(c) (i) Find the binomial expansion of $\frac{1+4x}{(1+x)(1+3x)}$ up to the term in x^3.

(3)

(ii) Find the range of values of x for which the binomial expansion of $\frac{1+4x}{(1+x)(1+3x)}$ is valid.

(2)

(Total 12 marks)

Q2.

(a) (i) Obtain the binomial expansion of $(1 - x)^{-1}$ up to and including the term in x^2.

(2)

(ii) Hence, or otherwise, show that

$$\frac{1}{3-2x} \approx \frac{1}{3} + \frac{2}{9}x + \frac{4}{27}x^2$$

for small values of x.

(3)

(b) Obtain the binomial expansion of $\frac{1}{(1-x)^2}$ up to and including the term in x^2.

(2)

(c) Given that $\frac{2x^2-3}{(3-2x)(1-x)^2}$ can be written in the form $\frac{A}{(3-2x)} + \frac{B}{(1-x)} + \frac{C}{(1-x)^2}$, find the values of A, B and C.

(5)

(d) Hence find the binomial expansion of $\frac{2x^2 - 3}{(3 - 2x)(1 - x)^2}$ up to and including the term in x^2.

(3)
(Total 15 marks)

Q3.

(a) The expression $(1 - 2x)^4$ can be written in the form

$$1 + px + qx^2 - 32x^3 + 16x^4$$

By using the binomial expansion, or otherwise, find the values of the integers p and q.

(3)

(b) Find the coefficient of x in the expansion of $(2 + x)^9$.

(2)

(c) Find the coefficient of x in the expansion of $(1 - 2x)^4(2 + x)^9$.

(3)
(Total 8 marks)

Q4.

(a) Obtain the binomial expansion of $(1 - x)^{-3}$ up to and including the term in x^2.

(2)

(b) Hence obtain the binomial expansion of $\left(1 - \frac{5}{2}x\right)^{-3}$ up to and including the term in x^2.

(2)

(c) Find the range of values of x for which the binomial expansion of $\left(1 - \frac{5}{2}x\right)^{-3}$ would be valid.

(2)

(d) Given that x is small, show that $\left(\frac{4}{2 - 5x}\right)^3 \approx a + bx + cx^2$, where a, b and c are integers.

(2)
(Total 8 marks)

Q5.

(a) (i) Using the binomial expansion, or otherwise, express $(2 + x)^3$ in the form

$8 + ax + bx^2 + x^3$, where a and b are integers.

(3)

(ii) Write down the expansion of $(2 - x)^3$.

(2)

(b) Hence show that $(2 + x)^3 - (2 - x)^3 = 24x + 2x^3$.

(2)

(c) Hence show that the curve with equation

$y = (2 + x)^3 - (2 - x)^3$

has no stationary points.

(3)
(Total 10 marks)

Q6.

(a) Using the binomial expansion, or otherwise, express $(1 + x)^4$ in ascending powers of x.

(3)

(b) (i) Hence show that $(1 + \sqrt{5})^4 = 56 + 24\sqrt{5}$.

(3)

(ii) Hence show that $\log_2(1 + \sqrt{5})^4 = k + \log_2(7 + 3\sqrt{5})$, where k is an integer.

(3)
(Total 9 marks)

Q7.

(a) Find the binomial expansion of $(1+x)^{-\frac{1}{2}}$ up to the term in x^2.

(2)

(b) Hence, or otherwise, obtain the binomial expansion of $\dfrac{1}{\sqrt{1+2x}}$ up to the term in x^2, in simplified form.

(3)

(c) Use your answer to part (b) with $x = -0.1$ to show that $\sqrt{5} \approx 2.23$.

(3)
(Total 8 marks)